# Chapter 6

# Matrices

## Only One Option Correct Type Questions

1. The number of 3 × 3 matrices $A$ whose entries are either 0 or 1 and for which the system $A\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ has exactly two distinct solutions, is **[IIT-JEE-2010 (Paper-1)]**

   (A) 0
   (B) $2^9 - 1$
   (C) 168
   (D) 2

2. Let $P = [a_{ij}]$ be a 3 × 3 matrix and let $Q = [b_{ij}]$, where $b_{ij} = 2^{i+j}a_{ij}$ for $1 \le i, j \le 3$. If the determinant of $P$ is 2, then the determinant of the matrix $Q$ is **[IIT-JEE-2012 (Paper-1)]**

   (A) $2^{10}$
   (B) $2^{11}$
   (C) $2^{12}$
   (D) $2^{13}$

3. If $P$ is a 3 × 3 matrix such that $P^T = 2P + I$, where $P^T$ is the transpose of $P$ and $I$ is the 3 × 3 identity matrix, then there exists a column matrix $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ such that **[IIT-JEE-2012 (Paper-2)]**

   (A) $PX = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$
   (B) $PX = X$
   (C) $PX = 2X$
   (D) $PX = -X$

4. Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix}$ and $I$ be the identity matrix of order 3. If $Q = [q_{ij}]$ is a matrix such that $P^{50} - Q = I$, then $\dfrac{q_{31} + q_{32}}{q_{21}}$ equals **[JEE (Adv)-2016 (Paper-2)]**

   (A) 52
   (B) 103
   (C) 201
   (D) 205

5. How many 3 × 3 matrices $M$ with entries from {0, 1, 2} are there, for which the sum of the diagonal entries of $M^T M$ is 5? **[JEE (Adv)-2017 (Paper-2)]**

   (A) 162
   (B) 135
   (C) 126
   (D) 198

6. Let $M = \begin{bmatrix} \sin^4\theta & -1-\sin^2\theta \\ 1+\cos^2\theta & \cos^4\theta \end{bmatrix} = \alpha I + \beta M^{-1}$, where $\alpha = \alpha(\theta)$ and $\beta = \beta(\theta)$ are real numbers, and $I$ is the 2 × 2 identity matrix. If $\alpha^*$ is the minimum of the set $\{\alpha(\theta) : \theta \in [0, 2\pi)\}$ and $\beta^*$ is the minimum of the set $\{\beta(\theta) : \theta \in [0, 2\pi)\}$, then the value of $\alpha^* + \beta^*$ is **[JEE (Adv)-2019 (Paper-1)]**

(A) $-\frac{17}{16}$

(B) $-\frac{29}{16}$

(C) $-\frac{31}{16}$

(D) $-\frac{37}{16}$

7. If $M = \begin{pmatrix} \frac{5}{2} & \frac{3}{2} \\ -\frac{3}{2} & -\frac{1}{2} \end{pmatrix}$, then which of the following matrices is equal to $M^{2022}$? **[JEE (Adv)-2022 (Paper-2)]**

(A) $\begin{pmatrix} 3034 & 3033 \\ -3033 & -3032 \end{pmatrix}$

(B) $\begin{pmatrix} 3034 & -3033 \\ 3033 & -3032 \end{pmatrix}$

(C) $\begin{pmatrix} 3033 & 3032 \\ -3032 & -3031 \end{pmatrix}$

(D) $\begin{pmatrix} 3032 & 3031 \\ -3031 & -3030 \end{pmatrix}$

## One or More Option(s) Correct Type Questions

8. For 3 × 3 matrices $M$ and $N$, which of the following statement(s) is (are) NOT correct?

**[JEE (Adv)-2013 (Paper-1)]**

(A) $N^TMN$ is symmetric or skew symmetric, according as $M$ is symmetric or skew symmetric

(B) $MN - NM$ is skew symmetric for all symmetric matrices $M$ and $N$

(C) $MN$ is symmetric for all symmetric matrices $M$ and $N$

(D) (adj $M$) (adj $N$) = adj($MN$) for all invertible matrices $M$ and $N$

9. Let $M$ be a 2 × 2 symmetric matrix with integer entries. Then $M$ is invertible if

**[JEE (Adv)-2014 (Paper-1)]**

(A) The first column of $M$ is the transpose of the second row of $M$

(B) The second row of $M$ is the transpose of the first column of $M$

(C) M is a diagonal matrix with nonzero entries in the main diagonal

(D) The product of entries in the main diagonal of $M$ is not the square of an integer

10. Let $M$ and $N$ be two 3 × 3 matrices such that $MN = NM$. Further, if $M \neq N$ and $M^2 = N^2$, then

**[JEE (Adv)-2014 (Paper-1)]**

(A) Determinant of $(M^2 + MN^2)$ is 0

(B) There is a 3 × 3 non-zero matrix $U$ such that $(M^2 + MN^2)U$ is the zero matrix

(C) Determinant of $(M^2 + MN^2) \geq 1$

(D) For a 3 × 3 matrix $U$, if $(M^2 + MN^2)U$ equals the zero matrix then $U$ is the zero matrix

11. Let $X$ and $Y$ be two arbitrary, $3 \times 3$, non-zero, skew-symmetric matrices and $Z$ be an arbitrary $3 \times 3$, non-zero, symmetric matrix. Then which of the following matrices is (are) skew symmetric?

**[JEE (Adv)-2015 (Paper-1)]**

(A) $Y^3Z^4 - Z^4Y^3$ (B) $X^{44} + Y^{44}$

(C) $X^4Z^3 - Z^3X^4$ (D) $X^{23} + Y^{23}$

12. Which of the following is(are) not the square of a $3 \times 3$ matrix with real entries?

**[JEE (Adv)-2017 (Paper-1)]**

(A) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

13. Let $S$ be the set of all column matrices $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ such that $b_1, b_2, b_3 \in \mathbb{R}$ and the system of equations (in real variables)

$$-x + 2y + 5z = b_1$$
$$2x - 4y + 3z = b_2$$
$$x - 2y + 2z = b_3$$

has at least one solution. Then, which of the following system(s) (in real variables) has (have) at least one solution for each $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \in S$?

**[JEE (Adv)-2018 (Paper-2)]**

(A) $x + 2y + 3z = b_1$, $4y + 5z = b_2$ and $x + 2y + 6z = b_3$

(B) $x + y + 3z = b_1$, $5x + 2y + 6z = b_2$ and $-2x - y - 3z = b_3$

(C) $-x + 2y - 5z = b_1$, $2x - 4y + 10z = b_2$ and $x - 2y + 5z = b_3$

(D) $x + 2y + 5z = b_1$, $2x + 3z = b_2$ and $x + 4y - 5z = b_3$

14. Let $M = \begin{bmatrix} 0 & 1 & a \\ 1 & 2 & 3 \\ 3 & b & 1 \end{bmatrix}$ and adj $M = \begin{bmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{bmatrix}$

where $a$ and $b$ are real numbers. Which of the following options is/are correct?

**[JEE (Adv)-2019 (Paper-1)]**

(A) $a + b = 3$

(B) $(\text{adj } M)^{-1} + \text{adj } M^{-1} = -M$

(C) If $M\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, then $\alpha - \beta + \gamma = 3$

(D) $\det(\text{adj } M^2) = 81$

15. Let $x \in \mathbb{R}$ and let $P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}, Q = \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 6 \end{bmatrix}$ and $R = PQP^{-1}$. Then which of the following options is/are correct? **[JEE (Adv)-2019 (Paper-2)]**

(A) $\det R = \det \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 5 \end{bmatrix} + 8$, for all $x \in \mathbb{R}$

(B) For $x = 1$, there exists a unit vector $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ for which $R\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

(C) For $x = 0$, if $R\begin{bmatrix} 1 \\ a \\ b \end{bmatrix} = 6\begin{bmatrix} 1 \\ a \\ b \end{bmatrix}$, then $a + b = 5$

(D) There exists a real number $x$ such that $PQ = QP$

16. Let $P_1 = I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$,

$P_3 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $P_4 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$,

$P_5 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, $P_6 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

and $X = \sum_{k=1}^{6} P_k \begin{bmatrix} 2 & 1 & 3 \\ 1 & 0 & 2 \\ 3 & 2 & 1 \end{bmatrix} P_k^T$

where $P_k^T$ denotes the transpose of the matrix $P_k$. Then which of the following options is/are correct?

**[JEE (Adv)-2019 (Paper-2)]**

(A) The sum of diagonal entries of $X$ is 18

(B) If $X\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \alpha\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, then $\alpha = 30$

(C) $X$ is a symmetric matrix

(D) $X - 30I$ is an invertible matrix

17. Let $M$ be a 3 × 3 invertible matrix with real entries and let $I$ denote the 3 × 3 identity matrix. If $M^{-1} = \text{adj}(\text{adj } M)$, then which of the following statements is/are ALWAYS TRUE?

**[JEE (Adv)-2020 (Paper-1)]**

(A) $M = I$

(B) $\det M = 1$

(C) $M^2 = I$

(D) $(\text{adj } M)^2 = I$

18. For any 3 × 3 matrix $M$, let $|M|$ denote the determinant of $M$. Let $I$ be the 3 × 3 identity matrix. Let $E$ and $F$ be two 3 × 3 matrices such that $(I - EF)$ is invertible. If $G = (I - EF)^{-1}$, then which of the following statements is (are) **TRUE**? **[JEE (Adv)-2021 (Paper-1)]**

(A) $|FE| = |I - FE|\,|FGE|$

(B) $(I - FE)(I + FGE) = I$

(C) $EFG = GEF$

(D) $(I - FE)(I - FGE) = I$

## Linked Comprehension Type Questions

**Paragraph for Q.Nos. 18 to 20**

Let $\mathscr{A}$ be the set of all 3 × 3 symmetric matrices all of whose entries are either 0 or 1. Five of these entries are 1 and four of them are 0. **[IIT-JEE-2009 (Paper-1)]**

**Choose the correct answer :**

19. The number of matrices in $\mathscr{A}$ is

(A) 12 (B) 6

(C) 9 (D) 3

20. The number of matrices $A$ in $\mathscr{A}$ for which the system of linear equations $A\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}1\\0\\0\end{bmatrix}$ has a unique solution, is

(A) Less than 4 (B) At least 4 but less than 7

(C) At least 7 but less than 10 (D) At least 10

21. The number of matrices $A$ in $\mathscr{A}$ for which the system of linear equations $A\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}1\\0\\0\end{bmatrix}$ is inconsistent, is

(A) 0 (B) More than 2

(C) 2 (D) 1

**Paragraph for Q.Nos. 22 to 24**

Let $p$ be an odd prime number and $T_p$ be the following set of 2 × 2 matrices

$$T_P=\left\{A=\begin{bmatrix}a & b\\ c & a\end{bmatrix}: a, b, c\in\{0, 1, 2, .......p-1\}\right\}$$ **[IIT-JEE-2010 (Paper-1)]**

**Choose the correct answer :**

22. The number of $A$ in $T_p$ such that $A$ is either symmetric or skew-symmetric or both, and det ($A$) divisible by $p$ is

(A) $(p-1)^2$ (B) $2(p-1)$

(C) $(p-1)^2+1$ (D) $2p-1$

23. The number of $A$ in $T_p$ such that the trace of $A$ is not divisible by $p$ but det ($A$) is divisible by $p$ is

[**Note :** The trace of a matrix is the sum of its diagonal entries]

(A) $(p-1)(p^2-p+1)$ (B) $p^3-(p-1)^2$

(C) $(p-1)^2$ (D) $(p-1)(p^2-2)$

24. The number of $A$ in $T_p$ such that det ($A$) is not divisible by $p$ is

(A) $2p^2$ (B) $p^3-5p$

(C) $p^3-3p$ (D) $p^3-p^2$

**Paragraph for Q.Nos. 25 to 27**

Let $a$, $b$ and $c$ be three real numbers satisfying

$$[a\quad b\quad c]\begin{bmatrix}1 & 9 & 7\\ 8 & 2 & 7\\ 7 & 3 & 7\end{bmatrix}=[0\quad 0\quad 0]\qquad ...(E)$$ **[IIT-JEE-2011 (Paper-1)]**

**Choose the correct answer :**

25. If the point $P(a, b, c)$, with reference to $(E)$, lies on the plane $2x + y + z = 1$, then the value of $7a + b + c$ is

(A) 0
(B) 12
(C) 7
(D) 6

26. Let $\omega$ be a solution of $x^3 - 1 = 0$ with $\text{Im}(\omega) > 0$. If $a = 2$ with $b$ and $c$ satisfying $(E)$, then the value of $\frac{3}{\omega^a} + \frac{1}{\omega^b} + \frac{3}{\omega^c}$ is equal to

(A) –2
(B) 2
(C) 3
(D) –3

27. Let $b = 6$, with $a$ and $c$ satisfying $(E)$. If $\alpha$ and $\beta$ are the roots of the quadratic equation $ax^2 + bx + c = 0$, then $\sum_{n=0}^{\infty}\left(\frac{1}{\alpha}+\frac{1}{\beta}\right)^n$

(A) 6
(B) 7
(C) $\frac{6}{7}$
(D) $\infty$

## Matrix-Match / Matrix Based Type Questions

28. Match the Statements/Expressions in **Column-I** with the Statements/Expressions in **Column-II**.

**[IIT-JEE-2008 (Paper-2)]**

| **Column-I** | **Column-II** |
|---|---|
| (A) The minimum value of $\frac{x^2+2x+4}{x+2}$ is | (p) 0 |
| (B) Let $A$ and $B$ be 3 × 3 matrices of real number, where $A$ is symmetric, $B$ is skew-symmetric, and $(A + B)(A - B) = (A - B)(A + B)$. If $(AB)^t = (-1)^k AB$, where $(AB)^t$ is the transpose of the matrix $AB$, then the possible values of $k$ are | (q) 1 |
| (C) Let $a = \log_3 \log_3 2$. An integer $k$ satisfying $1 < 2^{(-k+3^{-a})} < 2$, must be less than | (r) 2 |
| (D) If $\sin\theta = \cos\phi$, then the possible values of $\frac{1}{\pi}\left(\theta \pm \phi - \frac{\pi}{2}\right)$ are | (s) 3 |

## Integer / Numerical Value Type Questions

29. Let $M$ be a 3 × 3 matrix satisfying $M\begin{bmatrix}0\\1\\0\end{bmatrix}=\begin{bmatrix}-1\\2\\3\end{bmatrix}$, $M\begin{bmatrix}1\\-1\\0\end{bmatrix}=\begin{bmatrix}1\\1\\-1\end{bmatrix}$, and $M\begin{bmatrix}1\\1\\1\end{bmatrix}=\begin{bmatrix}0\\0\\12\end{bmatrix}$

Then the sum of the diagonal entries of $M$ is **[IIT-JEE-2011 (Paper-2)]**

30. Let $z = \frac{-1+\sqrt{3}i}{2}$, where $i = \sqrt{-1}$, and $r, s \in \{1, 2, 3\}$. Let $P = \begin{bmatrix} (-z)^r & z^{2s} \\ z^{2s} & z^r \end{bmatrix}$ and $I$ be the identity matrix of order 2. Then the total number of ordered pairs $(r, s)$ for which $P^2 = -I$ is **[JEE (Adv)-2016 (Paper-1)]**

31. For a real number $\alpha$, if the system $\begin{bmatrix} 1 & \alpha & \alpha^2 \\ \alpha & 1 & \alpha \\ \alpha^2 & \alpha & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ of linear equations, has infinitely many solutions, then $1 + \alpha + \alpha^2 =$ **[JEE (Adv)-2017 (Paper-1)]**

32. Let $P$ be a matrix of order $3 \times 3$ such that all the entries in P are from the set $\{-1, 0, 1\}$. Then, the maximum possible value of the determinant of $P$ is ____. **[JEE (Adv)-2018 (Paper-2)]**

33. The trace of a square matrix is defined to be the sum of its diagonal entries. If $A$ is a $2 \times 2$ matrix such that the trace of $A$ is 3 and the trace of $A^3$ is $-18$, then the value of the determinant of $A$ is

**[JEE (Adv)-2020 (Paper-2)]**

34. Let $\beta$ be a real number. Consider the matrix $A = \begin{pmatrix} \beta & 0 & 1 \\ 2 & 1 & -2 \\ 3 & 1 & -2 \end{pmatrix}$. If $A^7 - (\beta - 1)A^6 - \beta A^5$ is a singular matrix, then the value of $9\beta$ is _____.

**[JEE (Adv)-2022 (Paper-2)]**

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